

Aditya Hota

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EXPERIENCE

Internship

May 2025 - June 2025

Batoi Systems, Bhubaneswar

- Productionized and deployed a hackathon winning full-stack application. Built during Srusti Tech Hack Hackathon 2025, integrating Batoi Insight workflows into core platform services.
- Implemented and configured 4 recurring surveys with single and multi-stage forms, collecting user responses at scheduled intervals, and generating automated reports with 8+ metrics for administrators to monitor survey results.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend & Architecture: Node.js, Express.js, RESTful APIs, WebSockets, Redis, Microservices, Monorepo (Turborepo), LangChain, Git

Cloud & DevOps: Docker, Kubernetes, GitHub Actions (CI/CD), Trivy, Terraform, AWS, GCP, Cloudflare

Frontend & Database: PostgreSQL, Prisma ORM, HTML, React, Next.js, Tailwind CSS

ACHIEVEMENTS

- **Published Indian Patent:** *An Offline First, AI Enabled System for Decentralized Issue Reporting and Immutable Resolution Tracking*. Patent grant pending. Publication No: IN202631019043
- **Hackathon Winner:** Code Relay 2026 (IIT Bhubaneswar), Srusti Tech Hack Hackathon 2025.
- **Grand Finalist & Runner-up:** Smart India Hackathon (SIH) 2025, Founders Renaissance 2025 (IIIT Bhubaneswar), Google Solution Challenge 2026 (**top 106 out of 85000 teams**)
- **Project Lead, Web Development** at Google Developer Groups (GDG), C.V. Raman Global University. Leading workshops and web development initiatives on campus (October 2025 - Present).

EDUCATION

C. V. Raman Global University, Bhubaneswar, Odisha, India

July 2023 - July 2027

B.Tech, Computer Science and Engineering

CGPA: 8.35

PROJECTS

Swaraj Desk: Intelligent Civic Operations Platform | github.com/theogaditya/GSC-2026-Primeagen

Tech Stack: Next.js, Bun/Express, PostgreSQL, Prisma, Redis, LangChain, FastAPI, Docker, Kubernetes, WebSockets, GitHub Actions, Trivy, Grafana, Uptime Kuma

- Built a multi-department grievance platform for complaint tracking and management. With live community feed, **offline sync**, geo-tagging, and **20+ language** support. Implemented 4 role-based dashboards for agents, admins, and civic partners, enabling complaint assignment, resolution, escalation, and analytics.
- Developed **AI-assisted complaint processing pipeline** for automated classification, moderation, recommendations, and workload assignment. Built a bilingual chatbot with real-time user actions.
- Engineered **5 distributed microservices** with Redis queues. Deployed on Kubernetes with Ingress, GitHub Actions CI/CD, Trivy security scanning, and Grafana/Loki observability.

Nook: Zero-Knowledge Personal Cloud Vault | vault.adityahota99.workers.dev

Tech Stack: React, TypeScript, Vite, Cloudflare Workers, D1, R2, WebCrypto, Argon2id, WebAuthn, Tailwind CSS

- Designed a zero-knowledge cloud vault ensuring 100% of file content and metadata stay encrypted in client-side and **inaccessible to server and database**. With passkey only authentication. No email, password, or user name is collected or stored.
- Architected a **client-side trust model** using hierarchical encryption system (Argon2id, HKDF, and BIP-39 recovery) with browser-based key management. Implemented 5 MB **encrypted chunking** with streaming upload/download, multi-device sessions, rate limiting, and tamper detection.

PUBLICATION

Kubernetes Autoscaling for Stateful Workloads | github.com/theogaditya/Kubernetes-Custom-Controller

- **Authored 2 research papers**, under review at *Software: Practice and Experience* and *Concurrency and Computation: Practice and Experience*, on Kubernetes autoscaling for stateful workloads.
- Demonstrated failure of CPU-based Kubernetes HPA for persistent WebSocket workloads: **87.5%** over-provisioning, reconnection storms up to **1,400 conn/s**, and permanent connection loss.
- Developed StatefulAutoscaler, a custom Kubernetes controller using connection-aware scaling, sliding-window stabilization and bounded scale-down. Generalized to **MQTT**, achieving **44%** faster scale-up than HPA, and improved scale-up and scale-down efficiency.